Complex Hamiltonian learning and KAM-preserving magnetic control of a catheter model

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Anonymous authors

Abstract

We study how a learned energy can connect numerical integration, nonlinear stability and magnetic control of a clamped–free catheter model. Starting from the beam's complete magnetic potential, we derive a power-consistent complex Hamiltonian representation and specialize a classical fourth-order central symplectic scheme to its dynamics. Matched learning and integration comparisons separate representation, model and numerical errors. State-only COMSOL observations then provide nonzero-loss coefficient and analytic-error bounds, which establish nonlinear twist and support an a posteriori KAM proof for the same fitted Hamiltonian at approximately 1.9 mm amplitude. We derive magnetic force-and-torque feedback that realizes these conservative dynamics despite damping and lifts the invariant torus to all 64 beam modes. The resulting whole-centreline bound is below 1.971 mm for the specified startup. A separate input-work constraint guarantees clearance during driven motion; an S-shaped endpoint task achieves $6.42\,\mu$m RMS tracking error. Complete proofs and independent computations connect the learned function to these model-level operating bounds under explicit sensing and actuation assumptions.

Keywords

magnetic catheter robots, complex Hamiltonian learning, symplectic integration, KAM theory, magnetic control

1 Background and research question

Magnetic catheter robots combine distributed elastic deformation with remote actuation. A field rotates a magnetic segment through torque, while a field gradient can also exert force; the resulting motion depends on the mechanical boundary conditions and on dissipation throughout the flexible body (Abbott et al. 2020; Kim et al. 2019). For a catheter moving in a confined space, predicting the tip trajectory is only part of the problem. The controller must also respect the motion of the intervening centreline. We investigate how a learned dynamical model can support both recurrent, bounded motion and an explicit clearance condition for a magnetically actuated cantilever.

Reduced-gravity operation motivates this study. Long-duration spaceflight presents distinctive medical challenges, including the fluid redistribution associated with spaceflight-associated neuro-ocular syndrome (NASA n.d.). These concerns motivate research on remotely actuated devices, but do not determine a catheter's material parameters, surrounding flow or boundary conditions. We therefore formulate a mechanical problem that can be tested independently of a clinical scenario: a clamped–free elastic beam with a magnetic tip, viscous losses and specified magnetic inputs. The computational model is planar and assumes small slopes. Its parameters describe an illustrative device, and its independent finite-element computations test that stated model. Three-dimensional anatomy, contact and in-orbit operation remain separate application questions.

The central question is whether the *same learned energy* can support a structure-preserving numerical method, a rigorous nonlinear stability result and a realizable magnetic command. These are connected requirements. An accurate predictor may accumulate numerical energy error; an energy-conserving predictor may lack the nonlinear nondegeneracy required by KAM theory; and an invariant torus of a reduced conservative model may disappear under the damping present in the physical plant. Addressing each requirement with a different oscillator would leave the connections unresolved. We instead retain the beam's complete magnetic potential and a fixed dimensional map throughout learning, analysis and control.

1.1 From mechanical energy to learned dynamics

Variational rod mechanics supplies a consistent route from distributed elasticity to a finite-dimensional energy (Simo 1985; Antman 2005). In particular, the magnetic boundary moment must enter the mechanical work balance, and modal reduction must preserve the relation between configuration, momentum and velocity. These observations determine our model construction. The static bias field contributes to stored energy; modulation of the transverse field contributes input work. Dissipation is represented separately, so that learning a conservative residual cannot inadvertently change the sign of the dissipated power. The broader rod formulation and its force accounting are given in Supplement S1; the main text retains the specialization used in the proofs and experiments.

Hamiltonian neural networks learn a scalar energy and differentiate it to generate a vector field (Greydanus et al. 2019). This imposes the canonical structure directly and makes energy derivatives available for subsequent analysis. Dissipative SymODEN and port-Hamiltonian networks incorporate losses and external inputs (Zhong et al. 2020; Desai et al. 2021). We use the same separation because a magnetic catheter is driven and damped: the conservative energy determines its reversible motion, a nonnegative dissipation head represents losses, and magnetic ports account for supplied work. The complex-valued representation couples these quantities through shared features while retaining their distinct physical decoding.

Complex networks provide a natural notation for coupled modal position and momentum (Trabelsi et al. 2018). However, the representation alone does not establish a more accurate model. Approximation advantages proved for complex-reaction networks depend on their function classes, activations and comparison resources (Zhang et al. 2022); they cannot be transferred to an arbitrary smooth Hamiltonian architecture. The relevant requirement here is control of derivatives of a real energy on a specified domain. Smooth approximation results (Hornik 1991) motivate a finite analytic feature class, while the accuracy attained by its fitted coefficients is evaluated directly. Matched real and complex implementations consequently serve as a control for representation, rather than as evidence of an assumed complex-network advantage.

This choice also makes the learned model inspectable. The feature class contains the full trigonometric magnetic potential and a polynomial residual; the latter includes momentum-dependent terms when they are learned from data. Keeping those terms is essential. Removing them after fitting would change the function whose approximation error and invariant torus are being assessed, and would generally change the velocity-to-momentum relation used by the controller. Our analysis and feedback therefore differentiate the exported fitted Hamiltonian itself.

1.2 From a symplectic update to an invariant torus

Long-time prediction introduces a second issue: the numerical map should preserve the geometric structure of the conservative model. Generating functions provide a classical construction of canonical difference schemes (Feng et al. 1989), and symplectic integration distinguishes preservation of the symplectic form from exact conservation of energy (Hairer et al. 2006). We specialize the fourth-order central scheme of Feng et al. (1989, p. 95) to the learned Hamiltonian, deriving its derivative terms, local solvability, symplecticity and order conditions. The formula and its general geometric properties are classical; its explicit realization and evaluation for the fitted magnetic model form the numerical part of this work. This distinction matters particularly for a nonseparable learned Hamiltonian, for which a kinetic–potential splitting cannot simply be reused.

Numerical structure preservation still does not prove persistence of quasiperiodic motion. KAM theory requires a sufficiently regular Hamiltonian, suitable nonresonance and a nonlinear nondegeneracy condition (Pöschel 2001; Bounemoura 2020). In this problem, the magnetic cosine supplies a quartic contribution that can produce the required twist. Consequently, keeping only the linear frequencies would discard precisely the information needed to connect the mechanical model to the theorem. The learning analysis must control the fitted nonlinear coefficients and their analytic derivatives, not merely trajectory error on the training samples.

The relation between nonzero Hamiltonian-network training loss and KAM persistence has been studied by Chen et al. (2022). We build on that question by deriving bounds for a specified feature class and independently integrated state observations. The statistical argument explains what distributional training information can imply under stated sampling and regularity assumptions. A deterministic integral calculation then supplies the error budget used in the numerical proof. These two roles are kept explicit: a statistical loss estimate motivates the connection, while the finite-amplitude result uses verified bounds for the particular fitted function.

An a posteriori KAM theorem tests a candidate torus by its invariance defect and geometric nondegeneracy (Haro and Luque 2019). This permits a direct conclusion about the trained Hamiltonian, without substituting a convenient polynomial surrogate. We use that theorem to establish a millimetre-scale invariant torus and then relate its twist to a surrounding positive-measure family (Bounemoura 2016). The existence of this family does not prescribe a percentage over an arbitrary experimental domain. Quantitative measure estimates require a specified domain and additional constants (Biasco and Chierchia 2018; Chierchia and Koudjinan 2021); the complete local estimate is therefore presented with its domain in the Supplement.

A broader range of initial states is useful operationally, even when the available rigorous measure estimate is conservative. Weighted Birkhoff averages provide a complementary way to examine quasiperiodic behaviour over finite observation windows (Das et al. 2017; Duignan and Meiss 2023). We report their domain, observables, thresholds and step-size sensitivity. This experiment describes the computed trajectories, whereas the KAM proof establishes an infinite-time property of the certified invariant set. Keeping both results identifies what has been established mathematically and what the numerical exploration adds.

1.3 Contribution and progression of the paper

The contribution is the connection from an identified energy to a proved invariant motion and then to physical magnetic inputs on the same beam. It has three parts. First, we derive a power-consistent complex model and a fourth-order symplectic update, then distinguish representation effects, approximation error and integration error in their evaluation. Second, we obtain coefficient, analytic-error and twist bounds from nonzero-loss, state-only COMSOL identification and apply an external a posteriori theorem to that very Hamiltonian. Third, we derive a magnetic force-and-torque feedback that realizes its conservative low-mode dynamics despite damping, lift the certified torus to the complete finite-dimensional beam, and combine actuation with a whole-centreline clearance bound.

The resulting statements answer different parts of the mechanical problem. The invariant torus describes a persistent motion of the learned conservative system. The feedback construction specifies how that motion is realized in the damped plant under its sensing and actuator assumptions. The clearance inequality bounds the entire rod for admissible input work, including trajectories that need not lie on a KAM torus. Their combination gives concrete amplitude and magnetic-command limits. Independent COMSOL computations and matched tracking controls then test the implementation and identify where a learned controller is useful, without assuming that it must outperform a known-physics controller on every task.

Table 1 summarizes this progression. Section 2 derives the complex Hamiltonian and the symplectic difference scheme. Section 3 analyzes their approximation and numerical performance. Section 4 connects the fitted coefficients to KAM nondegeneracy and invariant tori. Section 5 turns those results into magnetic actuation and clearance conditions, and Section 6 draws the conclusions. Detailed force accounting, extended experiments and complete interval bounds accompany the main argument in the Supplement.

2 Derivation of the complex Hamiltonian and symplectic scheme

2.1 From magnetic rod mechanics to canonical variables

We specialize the distributed rod to a planar, small-slope magnetic cantilever, retaining its nonlinear tip potential. The transverse displacement $w(s,t)$ satisfies

$$\begin{aligned} \rho A_c w_{tt} + c_d w_t + EI w_{ssss} &= 0, \qquad 0 < s < L, \\ w(0,t) = w_s(0,t) &= 0, \qquad w_{sss}(L,t) = 0, \\ EI w_{ss}(L,t) &= \mu[B_y \cos\vartheta - B_0 \sin\vartheta], \end{aligned} \tag{1}$$

where $\vartheta = w_s(L,t)$, $A_c = \pi r_c^2$, and $I = \pi r_c^4/4$. The tip moment is the variation of $-\mu(B_0 \cos\vartheta + B_y \sin\vartheta)$; the free end carries a natural moment load, while its rotation remains unconstrained. The constant bias B_0 contributes to stored energy and the transverse field $B_y(t)$ supplies work. General spatial, fluid and contact terms are described in the supplementary mechanical formulation.

The computational beam has $L = 0.08\,\mathrm{m}$, $r_c = 0.0006\,\mathrm{m}$, $E = 5.0 \times 10^6\,\mathrm{Pa}$, $\rho = 1100\,\mathrm{kg\,m^{-3}}$, $\mu = 2.0 \times 10^{-5}\,\mathrm{A\,m^2}$, and $B_0 = 0.03\,\mathrm{T}$. Thirty-two consistent-mass cubic-Hermite elements leave 64 transverse displacement and rotation coordinates after clamping. With nodal vector q, consistent mass M, bending stiffness K, and tip-rotation selector e_ϑ, Galerkin projection of the weak beam equation gives the finite-element system. Two integrations by parts put the natural end moment on e_ϑ; the conservative energy follows from the same bending and

Table 1. Four steps connect the mechanical model to operation. Classical tools are cited where used; the last column identifies the model-specific work developed here.

Step	Established foundation	Work in this paper
Derive	Variational mechanics; central generating functions	Complex power decoder, magnetic feature class and full fourth-order update for the same energy.
Evaluate	Derivative approximation; convergence analysis	Matched learned models, integral identification and numerical error separation.
Establish persistence	Nondegenerate and a posteriori KAM theorems	Verified loss-to-coefficient bounds, nonlinear twist and a finite-amplitude torus of the fitted function.
Realize	Magnetic force/torque and energy balance	Two-port realization, full-model torus lift and clearance-constrained endpoint tracking.

magnetic potentials:

$$\begin{aligned} M\ddot{q} + \alpha M\dot{q} + Kq &= \mu e_\vartheta [B_y \cos\vartheta - B_0 \sin\vartheta], \\ H_0(q,p) &= \tfrac{1}{2} p^{\mathsf{T}} M^{-1} p + \tfrac{1}{2} q^{\mathsf{T}} K q \\ &\quad + E_0[1 - \cos(e_\vartheta^{\mathsf{T}} q)], \end{aligned} \tag{2}$$

where $p = M\dot{q}$, $E_0 = \mu B_0 = 6.0 \times 10^{-7}\,\mathrm{J}$, and the nominal damping rate is $\alpha = 0.55558524\,\mathrm{s}^{-1}$, with $c_d = \alpha\rho A_c$. The cosine potential remains exact within the small-slope beam model.

The bias-field stiffness is $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$. Choose $\Phi_n^{\mathsf{T}} M \Phi_n = I$ and $\Phi_n^{\mathsf{T}} K_b \Phi_n = \mathrm{diag}(\omega_j^2)$ for n retained modes, and write $q = \Phi_n a$, with reduced canonical momentum $\pi = \dot{a}$. Set $\omega_0 = \omega_1$, $\tau = \omega_0 t$, and $\nu_j = \omega_j/\omega_0$. Define the scalar position scales $S_j = \sqrt{E_0/(\omega_0\omega_j)}$. The balanced scaling is

$$a_j = \sqrt{\frac{E_0}{\omega_0\omega_j}}\, X_j, \qquad \pi_j = \sqrt{\frac{E_0\omega_j}{\omega_0}}\, Y_j. \tag{3}$$

Each position–momentum scale has product E_0/ω_0; together with H_0/E_0 and $\tau = \omega_0 t$, this gives canonical dimensionless equations. Thus $a_j = S_j X_j$. Define $S = \Phi_n\,\mathrm{diag}(S_j)$, $d = S^{\mathsf{T}} e_\vartheta$, and $\ell(s)^{\mathsf{T}} = N(s)S$, where $N(s)$ is the Hermite displacement row. Then $w(s) = \ell(s)^{\mathsf{T}} X$, $\vartheta = d^{\mathsf{T}} X$, and

$$\begin{aligned} h_*(X,Y) &= h_2(X,Y) + g(d^{\mathsf{T}} X), \\ h_2 &= \tfrac{1}{2}\sum_{j=1}^{n} \nu_j (X_j^2 + Y_j^2), \\ g(s) &= 1 - \cos s - \tfrac{1}{2} s^2. \end{aligned} \tag{4}$$

Subtracting $s^2/2$ avoids counting the bias stiffness twice. For the two-mode learning model, $\omega_0 \approx 11.417287283\,\mathrm{rad\,s^{-1}}$, $\nu \approx (1, 6.152117504)$, $d \approx (0.225670773, 0.321798806)^{\mathsf{T}}$, and $\ell(L) \approx (0.013506054, 0.005470926)^{\mathsf{T}}\,\mathrm{m}$. Displayed numerical values are rounded; calculations and verification use the full exported model coefficients. This chart is fixed before identification and is shared by learning, time integration and stability analysis.

2.2 Complex energy and a power-consistent decoder

Let $x = (X, Y) \in \mathbb{R}^{2n}$, $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, and

$$\begin{aligned} \zeta_j &= \frac{X_j + \mathrm{i}Y_j}{\sqrt{2}}, \qquad \{\zeta_j, \bar{\zeta}_k\} = -\mathrm{i}\delta_{jk}, \\ \partial_{\bar{\zeta}_j} &= \frac{\partial_{X_j} + \mathrm{i}\partial_{Y_j}}{\sqrt{2}}. \end{aligned} \tag{5}$$

A CHNN supplies a real energy h_η and a nonnegative Rayleigh readout $\mathcal{D}_\eta(X, v)$. Both are dimensionless after scaling physical dissipation by $1/(E_0\omega_0)$, so they can be represented as $\mathcal{H}_\eta = h_\eta + \mathrm{i}\mathcal{D}_\eta$. This paired readout has the physical decoder

$$\begin{aligned} X' &= v = \partial_Y h_\eta, \\ Y' &= -\partial_X h_\eta - \partial_v \mathcal{D}_\eta(X, v) + u\, d\cos(d^{\mathsf{T}} X). \end{aligned} \tag{6}$$

Here $u = B_y/B_0$, and $\partial_v \mathcal{D}_\eta$ is evaluated before substituting $v = \partial_Y h_\eta$. With $\mathsf{N} = \mathrm{diag}(\nu_j)$, the experiments use $\mathcal{D}_\eta = \widehat{\gamma} v^{\mathsf{T}} \mathsf{N}^{-1} v/2$, $\widehat{\gamma} \geq 0$; its physical value is α/ω_0. The chain rule gives

$$h_\eta' = -v^{\mathsf{T}} \partial_v \mathcal{D}_\eta + u\cos(d^{\mathsf{T}} X) d^{\mathsf{T}} v, \tag{7}$$

with an additional $\partial_\tau h_\eta$ if parameters vary in time. The damping term is nonpositive. In complex coordinates the same decoder reads

$$\zeta' = -\mathrm{i}\partial_{\bar{\zeta}} h_\eta + \frac{\mathrm{i}}{\sqrt{2}}[-\partial_v \mathcal{D}_\eta + u\, d\cos(d^{\mathsf{T}} X)]. \tag{8}$$

Thus the complex representation retains magnetic work and dissipation without treating them as an autonomous Hamiltonian perturbation. The symplectic scheme below concerns the conservative subsystem, $u = 0$ and $\mathcal{D}_\eta =$

0. Driven and damped computations retain their separate physical power balance.

2.3 Derivation of the fourth-order central scheme

Fix a real smooth learned Hamiltonian $h = h_\eta$ and write $f = J\nabla h$. We specialize the fourth-order central scheme of Feng et al. (1989, p. 95; Theorem 24), within the generating-function framework described by Hairer et al. (2006), to the complete learned magnetic energy. The following derivation retains its mixed position–momentum terms. Let Δ denote the dimensionless step and $m = (x_- + x_+)/2$ the endpoint midpoint. A central generating function $\mathcal{S}_\Delta$ defines a canonical map through

$$x_+ - x_- = J\nabla\mathcal{S}_\Delta(m). \tag{9}$$

Taking $\mathcal{S}_\Delta = \Delta h$ gives implicit midpoint. To determine the cubic correction, expand the exact trajectory about its temporal midpoint $x_0 = x(t + \Delta/2)$:

$$\begin{aligned} m &= x_0 + \tfrac{1}{8}\Delta^2 f' f(x_0) + O(\Delta^4), \\ x_+ - x_- &= \Delta f(x_0) \\ &\quad + \tfrac{1}{24}\Delta^3\{f''[f,f] + (f')^2 f\}(x_0) + O(\Delta^5). \end{aligned} \tag{10}$$

Since $x_0 = m - \Delta^2 f' f(m)/8 + O(\Delta^4)$, substitution yields

$$\begin{aligned} x_+ - x_- &= \Delta f(m) \\ &\quad + \frac{\Delta^3}{24}\{f''[f,f] - 2(f')^2 f\}(m) + O(\Delta^5). \end{aligned} \tag{11}$$

The negative coefficient comes from $1/24 - 1/8 = -2/24$; the temporal midpoint and endpoint midpoint are different at second order.

To recover a scalar generator, put $A = \nabla^2 h$. Since $f' = JA$ and $(f')^\mathsf{T} = -AJ$, differentiation gives

$$\begin{aligned} \nabla(f^\mathsf{T} A f) &= h^{(3)}[f,f] - 2AJAf, \\ J\nabla(f^\mathsf{T} A f) &= f''[f,f] - 2(f')^2 f. \end{aligned} \tag{12}$$

Consequently choose

$$\mathcal{S}_\Delta(x) = \Delta h(x) + \frac{\Delta^3}{24} f(x)^\mathsf{T}\nabla^2 h(x) f(x). \tag{13}$$

Equations (9) and (12) give the fourth-order difference scheme

$$\begin{aligned} x_+ &= x_- + g_\Delta(m), \\ g_\Delta &= \Delta f + \frac{\Delta^3}{24}[f''[f,f] - 2(f')^2 f]. \end{aligned} \tag{14}$$

Equivalently, $\zeta_+ - \zeta_- = -\mathrm{i}\partial_{\bar\zeta}\mathcal{S}_\Delta(m)$. All derivatives refer to one real scalar energy; the complex notation introduces no additional dynamical state.

Each step solves $r(x_+) = x_+ - x_- - g_\Delta(m) = 0$. Newton's correction is

$$x_+^{(k+1)} = x_+^{(k)} - \left[I - \tfrac{1}{2} D g_\Delta(m^{(k)})\right]^{-1} r(x_+^{(k)}). \tag{15}$$

The Jacobian uses energy derivatives through order four, evaluated analytically for the polynomial and cosine features. For sufficiently small Δ, it is invertible near the identity map. This construction applies to momentum-dependent learned energies, for which the physical kinetic–potential kick–drift split is generally unavailable.

3 Numerical and approximation analysis

3.1 Symplecticity, order and phase accuracy

For a differentiable local solution of (14), set $C_\Delta = J\mathcal{S}_\Delta''(m)/2$. Differentiating the implicit map gives

$$D\Psi_\Delta = (I - C_\Delta)^{-1}(I + C_\Delta). \tag{16}$$

The symmetry of $\mathcal{S}_\Delta''$ implies $C_\Delta^\mathsf{T} J + JC_\Delta = 0$, and the Cayley transform therefore satisfies $(D\Psi_\Delta)^\mathsf{T} J D\Psi_\Delta = J$. Moreover $\mathcal{S}_{-\Delta} = -\mathcal{S}_\Delta$, so interchanging the endpoints proves $\Psi_{-\Delta} = \Psi_\Delta^{-1}$. Equation (11) gives a local defect $O(\Delta^5)$, hence fourth-order global convergence on a fixed interval where derivatives are bounded and the implicit solution remains locally unique. These are properties of the classical generating-function construction (Feng et al. 1989, Theorems 17–18 and 23–24); see also Hairer et al. (2006). Here they apply to the identified magnetic energy without removing its momentum-dependent terms. Finite Newton residuals introduce additional numerical error, which is measured below.

The linear oscillator makes the phase improvement explicit. For $h = \nu(X^2 + Y^2)/2$, put $a = \nu\Delta + (\nu\Delta)^3/12$. The step is a rotation with eigenvalues $(1 \pm \mathrm{i}a/2)/(1 \mp \mathrm{i}a/2)$ and phase magnitude

$$\theta_\Delta = 2\arctan(a/2) = \nu\Delta - \frac{(\nu\Delta)^5}{120} + O(\Delta^7). \tag{17}$$

Table 2. Endpoint errors for the complete fitted GC17 energy. The state error is Euclidean; orders use successive step halvings.

Δ	Midpoint error (order)	Central4 error (order)
0.08	7.1351×10^{-2} (—)	1.8346×10^{-3} (—)
0.04	1.9383×10^{-2} (1.8802)	1.1784×10^{-4} (3.9606)
0.02	4.8966×10^{-3} (1.9849)	7.4149×10^{-6} (3.9902)
0.01	1.2265×10^{-3} (1.9972)	4.6422×10^{-7} (3.9976)

Its modulus is one for every real step. Implicit midpoint instead has $\theta_\Delta = \nu\Delta - (\nu\Delta)^3/12 + O(\Delta^5)$. Both preserve this oscillator's quadratic energy exactly, but their phase errors have different orders. For a nonlinear learned energy, symplecticity similarly distinguishes preservation of the canonical structure from exact preservation of the Hamiltonian value.

We compare the two schemes on the Python-trained, prior-informed complex model GC17 defined below, with all cosine and quartic coefficients held fixed. The initial state is $(0.1, 0.06, 0, 0.02)$ and the final dimensionless time is 10. An independent DOP853 reference uses relative and absolute tolerances 3×10^{-14} and 3×10^{-16}; its change under the preceding tolerance refinement is at most 8.32×10^{-14} per state component. Table 2 shows the predicted fourth- and second-order convergence.

The largest implicit residual is 2.00×10^{-15}, with at most three Newton evaluations per step. Differentiation of the converged step gives maximum symplectic defects 3.34×10^{-16} for Central4 and 2.24×10^{-16} for midpoint. A separate fourth-order finite-difference tangent agrees to 1.39×10^{-12} and has maximum symplectic defect 2.75×10^{-12}. The comparison establishes order and structure preservation at the measured solve accuracy. It does not compare the methods at equal computational cost, since Central4 evaluates higher derivatives. Full step records are given in supplementary Table S12.

3.2 Complex features and derivative approximation

The learned energy is real-valued in the complex coordinates $(\zeta, \bar{\zeta})$. Its quadratic part is $h_2 = \sum_j \nu_j \zeta_j \bar{\zeta}_j$; the magnetic feature can be written with complex exponentials and their conjugates. Products of complex affine forms and conjugate forms span the same real quadratic and quartic polynomials as real coordinates. For the two-mode model, let $m_4(x)$ contain the 35 monomials of total degree four, let v_4 be the coefficient vector of $-(d^{\mathsf{T}}X)^4/24$, and let the 34 columns of U span its orthogonal complement. We use

$$F(x) = \begin{pmatrix} g(d^{\mathsf{T}}X)/\|v_4\|_2 \\ U^{\mathsf{T}} m_4(x) \end{pmatrix},$$
$$h_\eta(x) = h_2(x) + \eta^{\mathsf{T}} F(x) = h_2 + a\, g(d^{\mathsf{T}}X) + m_4(x)^{\mathsf{T}} U c, \tag{18}$$

where $\eta = (a\|v_4\|_2, c)$. The full cosine is retained, its leading quartic direction is not duplicated, and the physical target has $a = 1, c = 0$. Polynomial multiplication is realizable by square activations through $uv = [(u+v)^2 - (u-v)^2]/4$, giving a finite analytic complex-feature network with a linear real readout. The learned energy is generally nonseparable because the residual contains momentum monomials.

Smooth approximation concerns derivatives as well as function values. On a compact polydisk strictly inside an analytic target's power-series convergence domain, its series truncations approximate the target and each fixed finite order of its derivatives. Their polynomials admit the complex feature realization just described. This is a representability statement; the accuracy of a fitted energy is determined by its coefficients and data. The L^2 approximation result for complex-reaction networks in Zhang et al. (2022) is a different statement and does not supply the derivative accuracy required here.

For an in-class target $h_* = h_2 + \eta_*^{\mathsf{T}} F$, define $B_r(D) = \max_{|\beta| \le r} \sup_{x \in D} \|\partial^\beta F(x)\|_2$. For conservative derivative labels, let ℓ be the mean over N observations of the squared Euclidean residual of each four-component vector, and ϵ_{rms} the corresponding label-error bound. With $DF \in \mathbb{R}^{35\times 4}$ and $G = N^{-1}\sum_i DF(x_i)DF(x_i)^{\mathsf{T}}$, the triangle inequality gives

$$\begin{aligned} \|h_\eta - h_*\|_{C^r(D)} &\le B_r(D)\|\eta - \eta_*\|_2 \\ &\le \frac{B_r(D)(\sqrt{\ell} + \epsilon_{\mathrm{rms}})}{\sqrt{\lambda_{\min}(G)}}, \end{aligned} \tag{19}$$

provided G is positive definite; the C^r norm here is the maximum over scalar coordinate derivatives. This bound follows because $\|JDF^{\mathsf{T}}(\eta - \eta_*)\|_2 = \|DF^{\mathsf{T}}(\eta - \eta_*)\|_2$. For an out-of-class target, a separate approximation defect must enter both the derivative-label budget and the final function error. Section 4 develops the verified loss-to-stability bounds for the identified models.

The experiments solve the linear readout problem by regularized least squares. For a nonlinear complex-parameter implementation, the real gradient $G_w = 2\partial_{\bar{w}}\mathcal{L}$ obeys the usual descent bound when $\mathrm{Re}\,\rho - (L/2)|\rho|^2 > 0$ and the

Table 3. Five paired contrasts of tip RMSE, using the same eight held-out trajectories for each of three training seeds. Positive prior contrasts favour G; positive representation contrasts favour C.

Contrast	Mean ± sample SD (mm)
R − GR	$(1.08\pm1.44)10^{-5}$
C − GC	$(1.08\pm1.44)10^{-5}$
R − C	$(-4.24\pm4.84)10^{-13}$
GR − GC	$(0.75\pm1.99)10^{-15}$
(GR − GC) − (R − C)	$(4.25\pm4.82)10^{-13}$

real gradient is L-Lipschitz on the traversed region. The proof and stationarity conditions are given in supplementary Section S10. Neither this optimization statement nor complex representability asserts a universal advantage over an equivalent real feature class.

3.3 Matched identification and long-time prediction

We separate representation choice from the mechanical prior. R and C denote real and complex features; G fixes the physical quadratic energy, whereas B fits its 10 coefficients. Thus GR and GC fit 35 real parameters, and BR and BC fit 45. Within each prior choice, the real and complex models use the same function class, observations and regularizer expressed in common real coordinates. All states use the fixed chart (3). Only G fixes its quadratic Hamiltonian in that chart.

For each seed in $\{17, 43, 91\}$, the two-mode Python Galerkin model generates 12 conservative trajectories of 4 s, with 129 observations per trajectory. Initial modal radii are uniform on $[0.03, 0.20]$ and $[0.03, 0.16]$, with independent uniform phases. States are noiseless; each vector-field component receives uniform noise in $[-10^{-6}, 10^{-6}]$. The quadratic regularizer is 10^{-14}. Eight independently initialized 16 s trajectories are held out in full. DOP853 tolerances are 2×10^{-10} and 2×10^{-12}; a hundredfold refinement changes the first test tip trajectory by at most 1.92×10^{-8} mm.

For the paired contrasts, abbreviate BR and BC as R and C. The differences R−GR and C−GC measure the effect of fixing the quadratic prior; R−C and GR−GC compare representations at each prior level. Their difference, (GR−GC)−(R−C), is the interaction. Each contrast is computed within a common training seed before taking its mean and sample standard deviation.

The mean tip RMSE is 2.28×10^{-5} mm for BR and BC and 1.20×10^{-5} mm for GR and GC. The mechanical prior improves two seeds and worsens the third, giving a positive mean contrast with substantial seed variation (Table 3). Representation and interaction contrasts are below the reference-solver refinement difference; the matched real and complex fits are numerically equivalent here. These three-seed summaries carry no significance claim. Supplementary Table S4 gives the model-level tip, slope and energy errors. GC estimates $a = 1.00326 \pm 0.00154$ with nonzero training residuals. The energy error compares the physical target energy along predicted and reference states; it is distinct from the drift of a model's own conserved energy. These small errors reflect an in-class target, accurate synthetic derivatives and controlled noise.

On a separate 200 s continuation, spanning 363 first-mode periods, the three GC fits have tip RMSEs of 1.38×10^{-4} to 4.03×10^{-4} mm and maximum modal phase error 4.19×10^{-4} rad. Their own-energy relative drift stays below 5.11×10^{-9}. The trajectories were integrated with DOP853 at tolerances 2×10^{-11} and 2×10^{-13}: this tests learned long-time prediction, separately from the Central4 convergence test. Supplementary Figure S3 gives the phase and Poincaré diagnostics. Finite-time agreement motivates, but does not replace, the invariant-torus analysis.

3.4 Dissipation and independent finite-element comparisons

The Rayleigh readout is identified from eight additional unforced, damped 5 s trajectories per seed, with the conservative coefficients held fixed. On six independent driven tests with field amplitudes 4–8 mT and frequency ratios 0.65–1.20, GC tip RMSEs are 8.83×10^{-5}, 1.37×10^{-4} and 6.40×10^{-5} mm. Closing the learned drag port raises the error to approximately 3.01 mm. The largest work-balance residual is 4.41×10^{-17} J. These tests establish the role of the dissipative decoder; some are free-motion stress cases beyond the corridor, while safe operation is treated in Section 5.

Independent COMSOL assembly reproduces the consistent-mass and bending matrices to relative Frobenius differences of 7.75×10^{-15} and 3.05×10^{-14}. A held-out full-beam run starts from rest and uses the nominal drag and a 6 mT sinusoid at $9.4449490256\,\mathrm{rad\,s^{-1}}$, with ramp $1 - \exp[-(t/0.25)^2]$ over 8 s. Against this trajectory, the exact-cosine 2-, 4- and 8-mode predictors have tip RMSEs 0.0156168, 0.00407422 and 0.00102484 mm; the complete 64-mode Python model has 0.00005614 mm RMSE. The learned two-mode GC model gives 0.0156171–0.0156177 mm. Its additional error against the exact two-mode response is only 1.81×10^{-6}–3.48×10^{-6} mm, so reduction error dominates this comparison. Supplementary

Table S6 retains the maximum errors and solver refinement results.

A separate full-FE COMSOL dataset provides four conservative training trajectories and two independent tests. Direct fitting of position–velocity increments, with no acceleration labels, gives $a = 1.0223248822$ and test tip RMSEs 0.14836 and $0.05140\,\mu\mathrm{m}$. The corresponding exact two-mode errors are 0.24172 and $0.27459\,\mu\mathrm{m}$: the fitted residual partly absorbs reduction and time-integration discrepancies. Supplementary Section S5.5 reports the stricter-solver comparison and distinguishes this state-only fit from identification using known-force labels. The short two-mode COMSOL experiment used for the verified learning-to-KAM bound in Section 4 is a separate dataset. Keeping these comparisons distinct separates representation, mechanical prior, dissipation, and model reduction from the accuracy of the symplectic time discretization.

State-only weak-integral fitting also tests unknown elasticity, magnetic strength and drag, observation noise, and omitted gradient-field and quadratic-fluid forces. All predictors retain four modes and use independent full-FE tests. In the soft-rod, positive-gradient setting, adding the missing physical features reduces tip RMSE from 182.988 to $9.829\,\mu\mathrm{m}$; with 0.5% state noise, it reduces 183.028 to $9.757\,\mu\mathrm{m}$. A freely fitted quadratic potential gives still smaller errors, 7.258 and $7.216\,\mu\mathrm{m}$, showing that prior constraints can limit absorption of model-reduction error. Further settings and an independent altered-physics COMSOL run appear in supplementary Section S5.7. On six one-second forecasts along one slowly drifting full-FE trajectory, causal adaptation using past states reduces mean tip RMSE from 387.483 to $9.867\,\mu\mathrm{m}$. These nonautonomous forecasts do not inherit the stationary KAM certificate.

4 Connection to KAM nondegeneracy and invariant tori

4.1 *From nonzero learning error to an analytic bound*

We consider the conservative two-mode Hamiltonian identified in the fixed canonical coordinates $x = (X_1, X_2, Y_1, Y_2)$. The quadratic prior h_2 is fixed, and the finite analytic readout is $h_\eta = h_2 + \eta^{\mathsf{T}} F$. The full-cosine target belongs to this class at $\eta_\star$. We distinguish $h_{\widehat{\eta}}$ from its exported function $\widehat{h}$, including decoder rounding. This setting separates two requirements: learning must control the relevant function norms, and the resulting Hamiltonian must satisfy the nondegeneracy and smallness conditions of a KAM theorem.

The learning-to-KAM argument of Chen et al. (2022, Theorem 11) motivates this connection. For a general smooth network, nonzero gradient loss gives a high-derivative bound only with additional regularity and sampling assumptions. For independent samples with density at least $m_0 > 0$, a bounded, parameter-Lipschitz hypothesis class and a uniform $W^{s,p}$ bound M_s on the gauge-fixed differences give, with probability at least $1 - \delta$,

$$\begin{aligned} \|h_\eta - h_\star\|_{C^\ell} &\le C\big(G_N + M_s^{a_{\mathrm{GN}}} G_N^{1-a_{\mathrm{GN}}}\big), \\ G_N &= C_P m_0^{-1/p} (\widehat{R}_N + \Gamma_N)^{1/p}, \\ a_{\mathrm{GN}} &= \frac{\ell - 1 + d/p}{s - 1}. \end{aligned} \tag{20}$$

Here d is the phase-space dimension, $p > d$, $s > \ell + d/p$, $\widehat{R}_N$ is the empirical gradient loss, and Γ_N is a uniform covering-number deviation bound at confidence $1 - \delta$. Supplementary Theorem S3 specifies the constants and proves the bound uniformly over data-selected parameters. For $\ell > 2(\tau + 1)$ and $\tau > 1$, the right-hand side plus the physical perturbation from a nondegenerate integrable Hamiltonian must be below the $c_K \gamma^2$ budget of Bounemoura (2020, Theorem A); the domain and frequency-map conditions are specified in Supplementary Corollary S4. Correlated trajectory observations are instead treated by the following deterministic argument.

COMSOL supplies 17 position–velocity observations on each of $m = 64$ two-mode trajectories. Let $c_j : [0, T] \to \mathbb{R}^4$ be their degree-eight fitted curves, with endpoints fixed to the exported observations, and let $Lx = J\nabla h_2(x)$. Define

$$\begin{aligned} D_j &= T^{-1} \int_0^T J\nabla F(c_j(t))\, dt, \\ y_j &= T^{-1} \left[c_j(T) - c_j(0) - \int_0^T L c_j(t)\, dt \right], \\ G_I &= (4m)^{-1} \sum_j D_j^{\mathsf{T}} D_j. \end{aligned} \tag{21}$$

The labels use states and feature integrals, with no acceleration reconstruction. Write r_I for the RMS residual $D_j\widehat{\eta} - y_j$ over all $4m$ scalar observations. Let ϵ_I bound, in the same normalization, the integrated physical defects $e_j = T^{-1} \int_0^T (\dot{c}_j - J\nabla h_\star(c_j))\, dt$.

Proposition 1. Coefficient and analytic error from integral observations. *If* $\lambda_{\min}(G_I) \ge \underline{\lambda} > 0$, *then*

$$\begin{aligned} \|\widehat{\eta} - \eta_\star\|_2 &\le b_I := \frac{r_I + \epsilon_I}{\sqrt{\underline{\lambda}}}, \\ \|\widehat{h} - h_\star\|_r &\le b_I B_F(r) + \delta_{h,\mathrm{dec}}(r). \end{aligned} \tag{22}$$

Table 4. Outward bounds for the same COMSOL-identified Hamiltonian. The loss is nonzero; the analytic and twist estimates include decoder rounding.

Quantity	Bound
Training MSE	$[3.6030, 3.6031]\,10^{-23}$
$r_I,\ \epsilon_I$	$\le 6.0026\,10^{-12},\ 6.8835\,10^{-12}$
$\lambda_{\min}(G_I)$	$\ge 2.9302\,10^{-7}$
b_I	$\le 2.3805\,10^{-8}$
Twist error	$\le 1.0377\,10^{-7}$
$\sigma_{\min}(A_\star)$	$\ge 4.9032\,10^{-4}$
$\|\widehat{h} - h_\star\|_{0.3}$	$\le 1.3310\,10^{-9}$

Here $\|\cdot\|_r$ is the supremum norm on the complex Cartesian polydisc $|x|_\infty < r$, $B_F(r)^2 = \sum_j \|F_j\|_r^2$, and $\delta_{h,\mathrm{dec}}$ bounds coefficient-export rounding.

Proof. Exact endpoint matching gives $y_j - D_j\eta_\star = e_j$. The triangle inequality in the normalized stacked observation norm therefore bounds $\sqrt{\underline{\lambda}}\,\|\widehat{\eta} - \eta_\star\|_2$ by $r_I + \epsilon_I$. Cauchy–Schwarz in feature space gives the analytic bound.

The physical defect is verified separately against the declared target; a small loss alone does not bound unknown physics. Arb interval integration includes the sine-series remainder and decoder rounding. The resulting analytic norm controls derivatives on every smaller polydisc through $\|\partial^\alpha(\widehat{h} - h_\star)\|_{r'} \le \alpha!(r - r')^{-|\alpha|}\|\widehat{h} - h_\star\|_r$. The observation and coefficient proofs are given in Supplementary Section S4.5.

4.2 Magnetic nonlinearity supplies and preserves twist

For $h_\star = h_2 + g(d^{\mathsf{T}}X)$, where $g(u) = 1 - \cos u - u^2/2$, the magnetic quartic is $-(d^{\mathsf{T}}X)^4/24$. Its angular average in $I_j = (X_j^2 + Y_j^2)/2$ has Hessian

$$A_\star = -\frac{1}{8}\begin{pmatrix} d_1^4 & 2d_1^2d_2^2 \\ 2d_1^2d_2^2 & d_2^4 \end{pmatrix}, \qquad \det A_\star = -\frac{3}{64}d_1^4d_2^4 \neq 0. \tag{23}$$

This follows from $\langle X_j^4\rangle = 3I_j^2/2$ and $\langle X_1^2X_2^2\rangle = I_1I_2$. Twist is nonsingular although indefinite. If B maps readout coefficients to effective quartic coefficients, angular averaging gives

$$\|A_{\widehat{h}} - A_\star\|_2 \le \sqrt{19}\big(\|B\|_2 b_I + \delta_{\mathrm{dec}}\big) < \sigma_{\min}(A_\star). \tag{24}$$

The factor $\sqrt{19}$ bounds the quartic-to-twist operator, including all momentum-dependent quartics. The strict inequality is verified for the actual identified function in Table 4.

The higher magnetic terms remain in the analysis: $\|g(d^{\mathsf{T}}X) + (d^{\mathsf{T}}X)^4/24\|_r \le (Dr)^6\cosh(Dr)/720$, with $D = \|d\|_1$. An exact quartic normalization, valid under fourth-order nonresonance, therefore yields $\nu\cdot I + \frac{1}{2}I^{\mathsf{T}}A_{\widehat{h}}I + O(r^6)$. The complete remainder bound and the resulting local KAM families near equilibrium are established in Supplementary Sections S4.1–S4.2. A finite physical amplitude requires an additional a posteriori check.

4.3 A validated torus of the same learned Hamiltonian

We apply Haro and Luque (2019, Theorem 3.1), using the explicit constants in Appendix B of arXiv:1711.05515v1. The following is its computer-assisted application to this identified magnetic model.

Theorem 2. Finite-amplitude persistence after integral learning. *The Hamiltonian in Table 4, defined by its exact exported binary coefficients and with cosine multiplier $a \simeq 0.9999996888354415$, admits a real analytic embedded Lagrangian torus $K_\infty : \mathbb{T}^2 \to \mathbb{R}^4$ satisfying*

$$J\nabla\widehat{h}(K_\infty) = DK_\infty\varpi, \qquad \varpi = \frac{w_1}{2\pi}(1, r_\omega), \qquad w_1 = \frac{49999589341}{50000000000}, \qquad r_\omega = \frac{1876278\phi + 592877}{304979\phi + 96369}, \qquad \phi = \frac{1+\sqrt{5}}{2}. \tag{25}$$

Angles have period one and time is dimensionless. The exact torus is within $2.0894\,10^{-41}$ of the computed Fourier embedding on the complex strip of width $1/80$. Its dimensional beam reconstruction attains a tip displacement of at least 1.8977904 mm and has centreline displacement at most 1.8977962 mm at every phase.

Proof. The bounded continued-fraction coefficients of r_ω give $|k\cdot\varpi| \ge (9/1750)|k|_1^{-1}$ for every nonzero integer k. Thus the weaker exponent 3 required by the chosen verification holds for all integer resonances. On the strip $\rho = 1/40$, the Fourier candidate has analytic invariance residual at most $2.7310\,10^{-77}$, including interpolation tails. Its inverse averaged torsion has norm below 80526. The geometric, derivative and domain estimates in Supplementary Section S4.3 give the theorem's residual threshold greater than $8.6086\,10^{-76}$; the verified smallness quotient is below $0.031724 < 1$. The cited theorem supplies the exact torus and embedding correction. Applying the same modal reconstruction with interval bounds proves the displacement estimates. All polynomial coefficients, including momentum quartics, and the complete cosine are retained in this verification.

The integral-loss and torus calculations use identical exported coefficients under the exact convention of Supplementary Section S4.3. The former establishes norm and twist control; the latter checks the extra finite-amplitude invariance condition. This rigorous chain concerns independently observed two-mode dynamics. Full-FE identification error and the controlled full-model lift are separate questions addressed in Section 5.

4.4 Positive-measure families and their domain of interpretation

The a posteriori iteration also preserves a bound $\|\langle T_\infty\rangle^{-1}\| \le 2\,10^5$ on the exact torus's averaged torsion. In an analytic symplectic tube, a convergent quadratic normalization gives

$$\widehat{h}\circ\Psi = e + \varpi\cdot I + \tfrac{1}{2} I^{\mathsf{T}}\overline{A} I + O(|I|^3), \qquad \det\overline{A}\neq 0. \tag{26}$$

Bounemoura (2016, Theorem 6, $\alpha = 1$, arXiv:1412.0509) then implies positive-measure families of analytic Lagrangian invariant tori around K_∞, with relative density tending to one as the transverse action box shrinks. Supplementary Corollary S8 proves the coordinate and torsion correspondence and extends this conclusion to all seven certified Hamiltonians.

For a specified domain $U \subset \mathbb{R}^4$ of positive Liouville volume, the fraction is $f_{\mathrm{KAM}}(U) = \mu_L(U\cap\mathcal{K}_h)/\mu_L(U)$, where $\mathcal{K}_h$ is the union of analytic Diophantine Lagrangian tori. The local family theorem gives no numerical fraction on a preassigned operating domain. An explicit fixed-tube calculation using Chierchia and Koudjinan (2021, Theorems 1 and 3) is retained in Supplementary Section S11; its extremely small transverse width quantifies the conservatism of those sufficient bounds. For the broader domain $I_1, I_2 \in [0.0032, 0.0072]$, with independently uniform actions and phases, weighted Birkhoff diagnostics give $1024/1024$ finite-time candidates after 8192 dimensionless time units (717.5 s). At step $1/64$, all satisfy the eight-digit consecutive-window test at $T = 2048$ and 4096; the six- and ten-digit checks agree. These classifications, their sampling uncertainty and refinement checks are reported in Supplementary Section S5.6; they are not an infinite-time KAM-volume bound. Under the magnetic realization, the lifted family retains its low-mode parameter measure, without implying positive 128-dimensional volume for the damped plant.

5 Guidance for magnetic actuation and clearance

The invariant torus of Section 4 is a conservative motion. To use it on the damped beam, a controller must reproduce the learned dynamics through available magnetic inputs and account for the remaining modes. We first derive that realization and its full-model lift, then use a separate physical energy bound to govern motions away from the torus.

5.1 Realizing the learned dynamics through two magnetic ports

Let a_j, v_j denote the mass-normalized positions and velocities of all 64 transverse finite-element modes. Their bias-field frequencies are ω_j, and their tip displacement and rotation coefficients are b_j, t_j. A transverse tip force F and a transverse field B_y give

$$\begin{aligned} \ddot{a}_j + \alpha\dot{a}_j + \omega_j^2 a_j &= b_j F + t_j U,\\ U &= E_0(\vartheta - \sin\vartheta) + \mu B_y\cos\vartheta,\\ \vartheta &= \sum_j t_j a_j. \end{aligned} \tag{27}$$

The complete modal coefficients define the computational plant; their relative kinetic and stiffness form discrepancies from the assembled nodal matrices are below 1.31×10^{-14} and 3.01×10^{-9}, respectively (Supplement S6.1).

For the first two modes, write $a_i = S_i X_i$ with the diagonal scales of Section 2. Because the learned h contains momentum quartics, velocity determines Y through the implicit relation $h_Y(X,Y) = \mathrm{diag}(S_i)^{-1} v_{1:2}/\omega_0$. If $h_{YY} \succ 0$, this relation is locally invertible. Differentiating it along $\dot{X} = \omega_0 h_Y$, $\dot{Y} = -\omega_0 h_X$ gives the desired physical acceleration and hence the feedback

$$\begin{aligned} a^{\mathrm{des}}_{i,tt} &= S_i\omega_0^2(h_{YX}h_Y - h_{YY}h_X)_i,\\ \begin{pmatrix} F\\ U\end{pmatrix} &= A_{\mathrm{mag}}^{-1}\left(a^{\mathrm{des}}_{1:2,tt} + \alpha v_{1:2} + \mathrm{diag}(\omega_1^2,\omega_2^2)a_{1:2}\right),\\ A_{\mathrm{mag}} &= \begin{pmatrix} b_1 & t_1\\ b_2 & t_2\end{pmatrix}. \end{aligned} \tag{28}$$

Substitution into (27) enforces the desired acceleration. Differentiating the velocity relation once more and using invertibility of h_{YY} then recovers both canonical equations. This derivation uses the complete fitted gradient and Hessian.

The force and moment in (28) are physically realized by the local field $B(r) = (B_0, B_y) + G(r - r_{\mathrm{tip}})$, where

$$B_y = \frac{U - E_0(\vartheta - \sin\vartheta)}{\mu\cos\vartheta}, \qquad G = \frac{F}{\mu}\begin{pmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{pmatrix}. \tag{29}$$

For the dipole $m = \mu(\cos\vartheta, \sin\vartheta)$, direct multiplication gives $Gm = (0, F)$ and $\|G\|_2 = |F|/\mu$. Symmetry and zero trace of G satisfy the local source-free curl and divergence constraints. The construction requires independent field and gradient authority, observed modal and tip states, and $|\vartheta| < \pi/2$; it does not assume independently actuated mechanical modes.

Theorem 3. Full-model realization of the certified torus. *Suppose the learned Hamiltonian has an analytic embedded two-torus with physical angular frequency $\Omega = 2\pi\omega_0\varpi$. Assume $h_{YY} \succ 0$ on a domain containing this torus with convex Y-fibres for each X, A_{mag} is invertible and $\alpha > 0$. If the lifted motion stays in the domain $|\vartheta| < \pi/2$ of the available commands, feedback (28)–(29) gives an embedded invariant two-torus of the complete finite-dimensional plant. For each low-mode orbit on that torus, higher-mode deviations from its unique quasiperiodic response decay exponentially along closed-loop trajectories that remain in the stated command domain.*

Proof. The feedback reproduces the low-mode Hamiltonian exactly. Its demands F, U therefore depend only on the torus phase. With Fourier convention $e^{ik\cdot\phi}$, each higher-mode response is

$$a^*_{j,k} = \frac{b_j F_k + t_j U_k}{\omega_j^2 - (k\cdot\Omega)^2 + i\alpha(k\cdot\Omega)}, \qquad j \geq 3. \tag{30}$$

The denominator is nonzero and its inverse is bounded on real frequencies: at zero its real part is positive, and away from zero its imaginary part is nonzero. Analytic forcing thus gives an analytic response on a possibly smaller strip. Positive definiteness on each convex Y-fibre makes $Y \mapsto h_Y(X, Y)$ strictly monotone and injective. The physical low-mode coordinates therefore preserve the torus embedding. Differences between high-mode solutions satisfy homogeneous damped oscillator equations and decay exponentially.

Theorem 3 is the model-specific lift derived here; its input torus is obtained by applying the external KAM theorem in Section 4. With period-2π phases, $\Omega = 2\pi\omega_0\varpi$, approximately $(11.41719351, 70.24034116)$ rad/s. High-mode contraction leaves the conservative low-mode motions intact, so it does not make one selected torus attracting. The KAM-family measure concerns the four-dimensional conservative subsystem, not the volume of a two-torus in the complete 128-dimensional state space. In contrast, the unforced physical beam with positive damping has a globally asymptotically stable equilibrium (Supplementary Proposition S2); sustained recurrent motion requires the supplied magnetic work.

Table 5. Rigorous bounds for the COMSOL-integral learned model. Startup means zero higher-mode positions and velocities, with the low state on the certified torus.

Quantity	Upper bound
Centreline on response graph	1.922696 mm
Centreline including startup	1.970263 mm
Transverse field including startup	0.921665 mT
Gradient norm including startup	0.02572538 T/m

For the COMSOL integral fit, interval bounds give $h_{YY} \succeq 0.9999999983I$ on the real 0.3 box and exclude zero from $\det A_{\mathrm{mag}}$. The response and startup estimates in Supplement S6.2 include all 62 higher modes and the complete nonlinear forcing remainder. Cubic-Hermite spatial bounds yield the whole-centreline and command limits in Table 5. With the low state on the exact certified torus and all higher positions and velocities initially zero, the slope remains below 0.155261 rad. Hence $\cos\vartheta > 0.987971$, closing the feedback-domain condition for every phase and all time.

The full-state implementation is checked separately over 12 s (Figure 1). Its low-mode state differs from an independent integration of the learned Hamiltonian by at most 1.54×10^{-10} in canonical coordinates. Online COMSOL feedback, including its algebraic velocity-to-momentum inverse, is also tested: reducing its fixed step from 1 to 0.125 ms lowers the four-second tip RMSE against the independent full-state solver from 147.852 to 3.848 μm. The corresponding maximum differences fall from 355.845 to 8.658 μm. These convergence measurements test implementation accuracy; the all-time bounds in Table 5 come from the analytic and interval estimates.

5.2 Clearance beyond the invariant torus

Tracking a prescribed path need not keep the state on a conservative torus. We therefore derive a clearance condition from the physical energy itself. For the clamped small-slope beam, $w(s) = \int_0^s (s - \xi) w_{ss}(\xi)\, d\xi$. Since the static

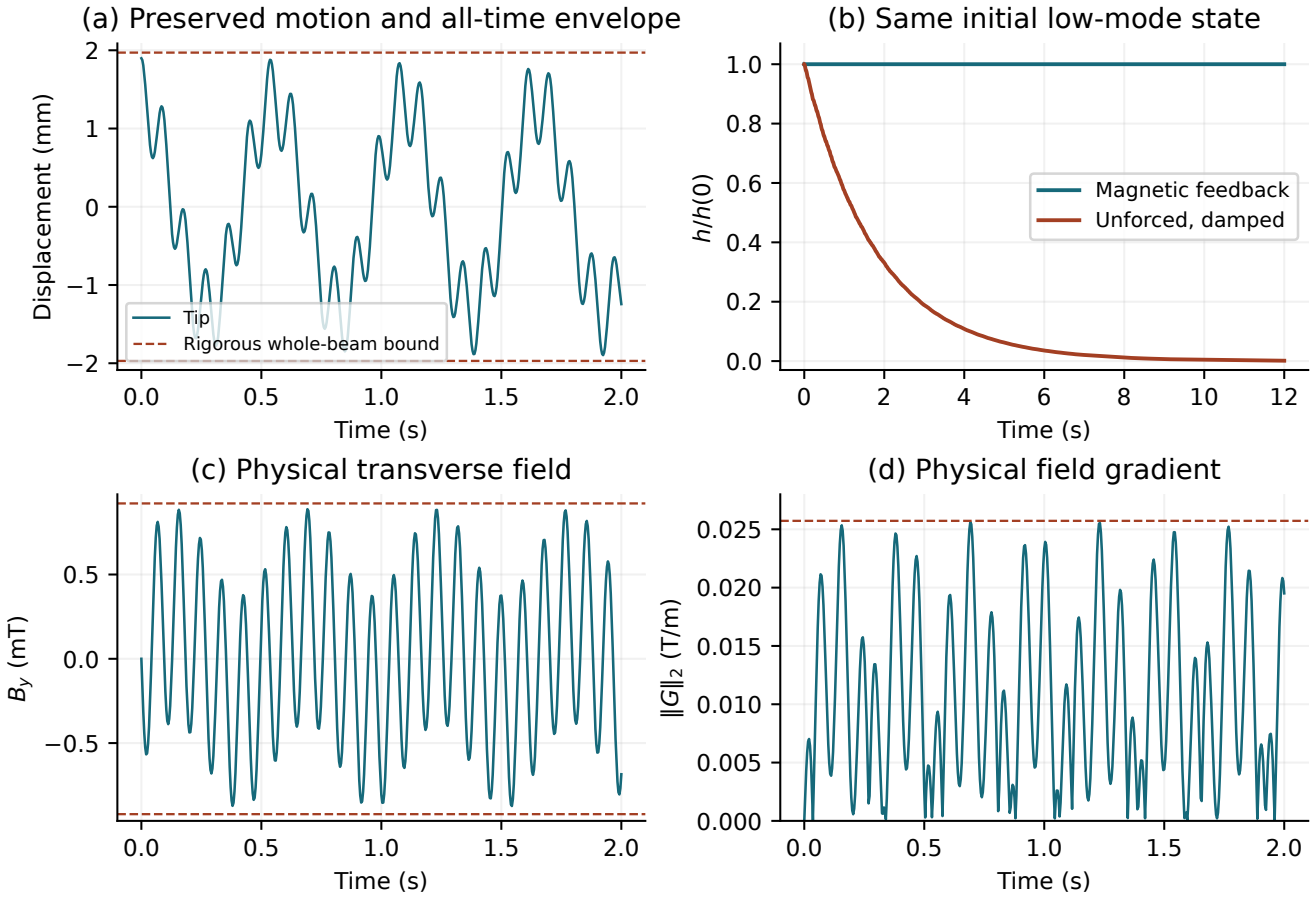


Figure 1. Realizing the certified learned dynamics on all 64 modes. The higher modes start at zero. Dashed envelopes include the startup transient and bound every phase and centreline point. Magnetic field and gradient jointly supply the force and torque; the passive comparison uses the same initial state without this feedback. Panels (a,c,d) show 2 s and panel (b) shows 12 s.

magnetic energy is nonnegative, Cauchy–Schwarz gives

$$|w(s,t)|^2 \le \frac{2s^3E(t)}{3EI}, \qquad E_c = \frac{3EIc^2}{2L^3}, \tag{31}$$

where c is the corridor half-width less the rod radius. Thus $E < E_c$ guarantees clearance of the entire centreline and its radius tube. This argument applies to sufficiently regular continuum solutions satisfying the power identity and separately to conforming Hermite finite-element solutions.

For a single transverse magnetic input, its actual power is a_pB_y, with $a_p = \mu\cos\vartheta\,\dot{\vartheta}$. Starting from a valid bound $\overline{E}_0 \ge E(0)$, let $\dot{W} = a_pB_y$ and $W(0) = \overline{E}_0$. Nonnegative dissipation gives $E \le W$. Choose $\overline{E}_0 < E_{\max} < E_c$, and project the requested field onto

$$|B_y| \le B_{\max}, \qquad a_pB_y \le \kappa(E_{\max} - W), \quad \kappa > 0. \tag{32}$$

Zero field is feasible while $W \le E_{\max}$. Moreover, (32) implies $E_{\max} - W(t) \ge (E_{\max} - \overline{E}_0)e^{-\kappa t}$. Together with (31), this proves the all-time clearance bound under exact port measurement and command application. Sensor errors, delays or additional work ports require corresponding budgets. This energy argument is independent of the KAM theorem; the two results cover different operating motions.

For the stated 4 mm net corridor, $E_c = 23.8565$ nJ. Taking the energy safety factor $E_{\max}/E_c = 0.95$ gives a continuous-time centreline bound of 3.898718 mm. A 10 mT sinusoidal request produces a 5.14478 mm sampled peak without the governor and a 3.14362 mm peak with it. Online COMSOL implementation gives 3.143643 mm and checks the same port-work inequality. The complete power derivation, open-loop field-variation bound and refinement data appear in Supplement S7.1.

5.3 Endpoint tracking and practical scope

To test a spatial task, axial translation of the clamped base is combined with magnetic bending. The tip advances 48 mm during a 12 s S-shaped stroke with lateral extrema ±1.5 mm. Axial acceleration ends before bending begins, and the stated model has no axial drag or guide contact. The base kinetic energy and its work are accounted for separately; the transverse governor still bounds clearance in the straight corridor (Supplement S7.5). A single uniform magnetic field

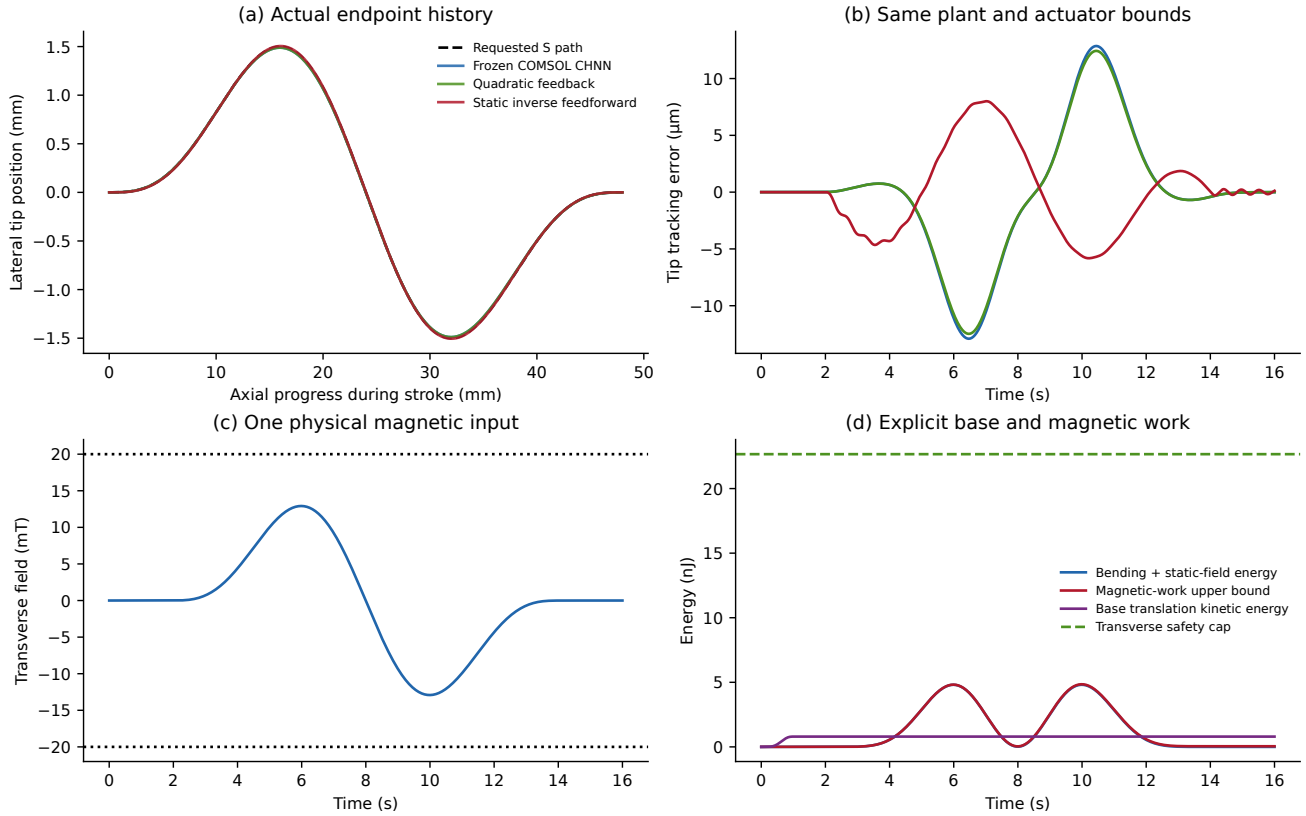


Figure 2. S-shaped endpoint tracking through axial base feed and transverse magnetic actuation. The same full beam and input limits are used for the matched controls. Base work is included in the energy balance. The known-physics inverse achieves the smallest error on this slow nominal task.

Table 6. Matched S-path controls. RMS errors use the 12 s stroke; maxima use the complete 16 s simulation.

Control	RMS (μm)	Max. (μm)
CHNN, tip feedback	6.419	12.907
Quadratic, tip feedback	6.207	12.476
Static inverse feedforward	4.124	8.005
CHNN, modal-output feedback	25.308	41.544

does not prescribe an arbitrary static S-shaped centreline: the evaluated task is the history of the moving endpoint.

The controller uses the full-FE COMSOL-trained Hamiltonian, its complete gradient and Hessian, observed low-mode projections and measured tip feedback. Its coefficients stay fixed during this experiment. Table 6 compares controls with the same trajectory, plant and 20 mT field limit. The learned controller achieves $6.419\,\mu$m RMS error; the known-physics static inverse is more accurate on this slow nominal path. Thus the example demonstrates feasible use of the learned model and the importance of the measured output, while retaining the stronger baseline result.

The learned command peaks at 12.9177 mT and the centreline at 1.48892 mm. Its magnetic-work bound is 4.84997 nJ, so the governor remains inactive for this feasible request. A separate COMSOL replay of the prescribed field agrees with the computed tip to within 4.038×10^{-10} m; this replay tests the mechanical response, whereas the preceding torus experiment recomputes the controller online. Figure 2 shows the resulting endpoint history.

These results give usable model-level guidance: preserve the full learned derivatives when computing magnetic commands, verify low-mode actuation rank and momentum invertibility, and bound total input work when leaving the certified invariant set. The field and gradient budgets assume available local magnetic authority rather than a calibrated coil design. Full-state projections and tip measurements are available in the simulations. Extending the same guarantees to partial sensing, large rotations, flowing fluid or anatomical walls requires those effects in the model and error budget. The demonstrated proofs and computations concern the specified beam, not a physical, clinical or in-orbit catheter experiment.

6 Conclusion

We connected complex Hamiltonian learning, symplectic integration and KAM-preserving magnetic control on one clamped–free beam. The derivation specializes a classical fourth-order central scheme to the fitted energy, while the

approximation analysis converts nonzero identification error into analytic and twist bounds. An external a posteriori theorem then establishes a finite-amplitude torus of that function. Our magnetic realization and full-model lift reproduce its dynamics despite damping, with explicit centreline and actuator envelopes.

Input-work control additionally guarantees clearance during driven tracking. The S-path tests demonstrate feasibility and retain the more accurate known-physics baseline; matched real and complex fits likewise show no intrinsic accuracy advantage from representation alone. The contribution lies in connecting the identified structure to proved motion and realizable inputs. Partial sensing, uncertain actuation and three-dimensional interaction require further models and error bounds.

Data and code availability

The accompanying package contains both article and supplement sources, native COMSOL models and construction scripts, exported matrices and trajectories, learning code, fitted coefficients, figure sources and interval-arithmetic records. The full-FE training observations and the two-mode integral observations used for the learning-to-KAM proof are identified separately.

Declarations

This work reports mathematical analysis and computer-generated mechanical data. No human, animal or in-orbit experiment was performed.

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